Cosmology

Problem Set 2: General Relativity

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These are the end of chapter problems for Chapter 2 of the notes. Equation numbers refer to the full notes.

1. The Big Bang

Starting with Friedmann 2,

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P)$$

Argue that the **strong energy condition** ($\rho + 3P \geq 0$) means that $a(t)$ is concave, and therefore that if we run the expansion backwards, a must reach zero at some finite time in the past, at most $1/H_0$ ago. Call that time $t = 0$. This is called the **the Big Bang**! Does the scale factor vanishing necessarily mean that space has 0 volume?

2. Matter And Dark Energy

A flat universe, containing matter and a cosmological constant, expands according to the Friedmann equation

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\left(\rho_\Lambda + \frac{\rho_0}{a^3}\right)$$

Solve this to find

$$a(t) = \beta(\sinh \alpha t)^{2/3}$$

with α and β constant. [Hint: you may find the substitution $b = a^{3/2}$ helpful.] Verify the expected asymptotic behaviour for early and late times.

Our universe today is flat, with $\Omega_m \approx 0.3$ and $\Omega_\Lambda \approx 0.7$ and $H_0^{-1} \approx 14$Gyr. Determine:

- The age of the universe.

- The age at which the expansion of the universe first started to accelerate.

- The age at which the energy density in matter and the cosmological constant were equal.

A cosmologically ignorant civilization measures $H_0^{-1} \approx 14$ Gyr but refuses to countenance the possibility of a cosmological constant. How old would they believe their universe to be if they assumed it was flat with $\Omega_m = 1$?

3. Einstein's Biggest Blunder

Einstein introduced the cosmological constant into his field equations to avoid the conclusion that the universe is expanding. In this problem, you will see that this was misguided.

a). Show that for a perfect fluid with positive density and pressure there is no static solution to the Einstein equations.

b). Consider now a universe with pressureless matter and a cosmological constant. Show that it is possible to obtain a static solution-called the Einstein static universe-if

$$\Lambda = 4\pi G\rho_m$$

What is the spatial curvature of this solution?

c). Show that the Einstein static universe is unstable to small perturbations. Hint: Consider a small perturbation around the static solution

$$\rho_m(t) = \rho_{m,0}[1 + \delta(t)]$$
$$a(t) = 1 + \epsilon(t)$$

with $|\delta| \ll 1$ and $|\epsilon| \ll 1$. Show that the perturbations δ and ϵ are related to each other and that they grow exponentially with time.

4. Dark Stars

With the arrival of the Einstein field equations, people were eager to begin solving it to find whatever solutions that they could find. We will consider the most symmetric and static object that we can solve for. Consider a spherically symmetric, static object with the metric

$$ds^2 = -g_{00}(r)dt^2 + g_{rr}(r)dr^2 + r^2 d\Omega^2$$

Using the Einstein equations

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0$$

a). Solve for the functions $g_{00}(r)$ and $g_{rr}(r)$ (you may use EinsteinPy if you wish. Or you can do it by hand).

b). The solution for your metric will admit 2 singularities. Using the scalar curvature R, determine which is a true singularity and which is an artifact of our coordinate system

c). At the coordinate singularity, determine what happens to measurements of time and measurements space. What happens to these measurements below the "event horizon"?

d). Once you have passed the event horizon, show that in order for you to leave this object, your velocity would need to be greater than the speed of light. It should be fairly obvious by now, but do you know what object you have just discovered?

5. The 3-Sphere

Show that the embedding $X_0 = \cos\chi$, $X_1 = \sin\chi\cos\theta$, $X_2 = \sin\chi\sin\theta\cos\phi$, $X_3 = \sin\chi\sin\theta\sin\phi$ describes the unit 3-sphere $\sum_i X_i^2 = 1$, and that the induced metric $\sum_i dX_i^2$ is $d\chi^2 + \sin^2\chi\, d\Omega^2$. What is the total volume of the 3-sphere of radius a? (This would be the volume of a closed universe with $k = +1$.)